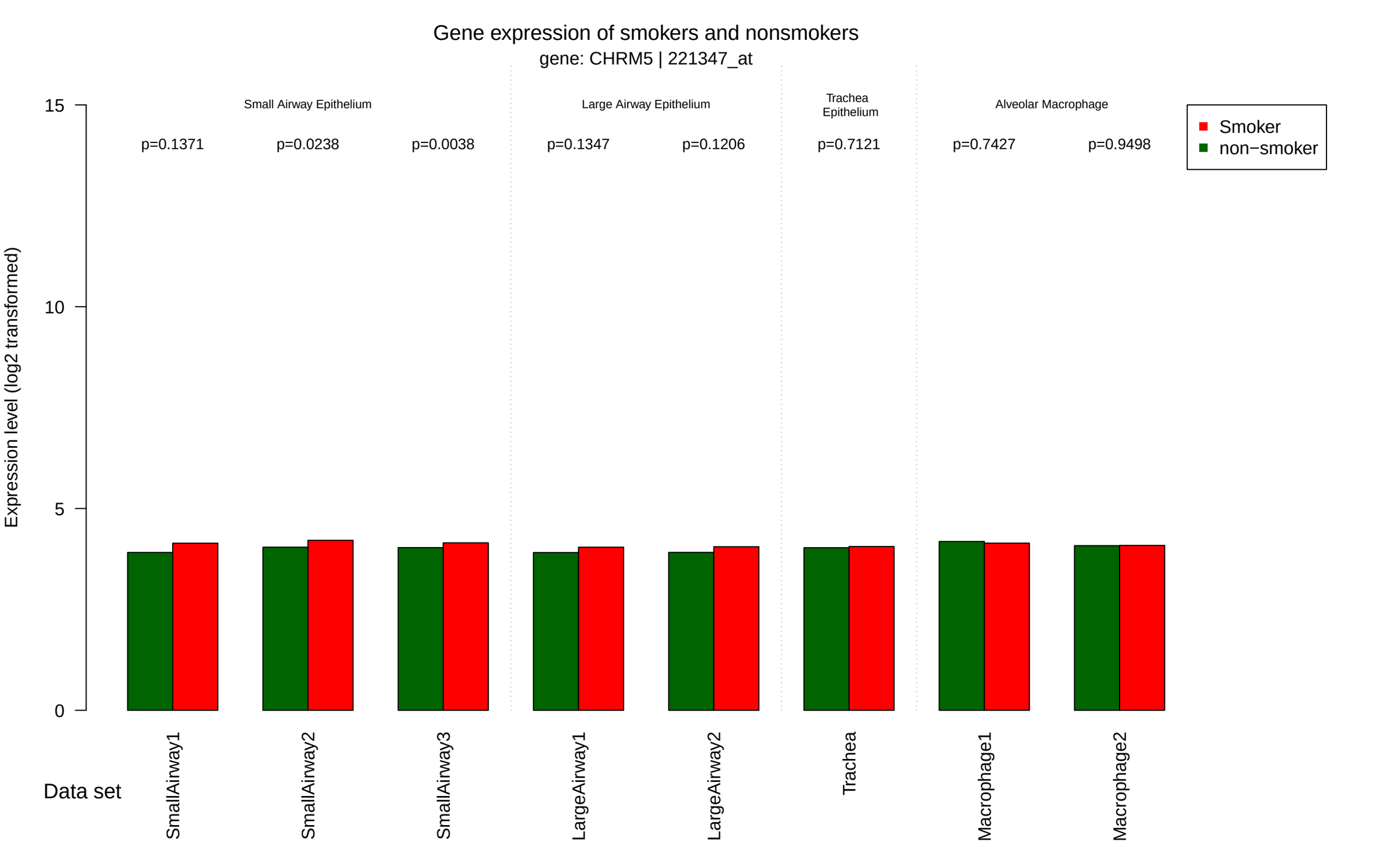
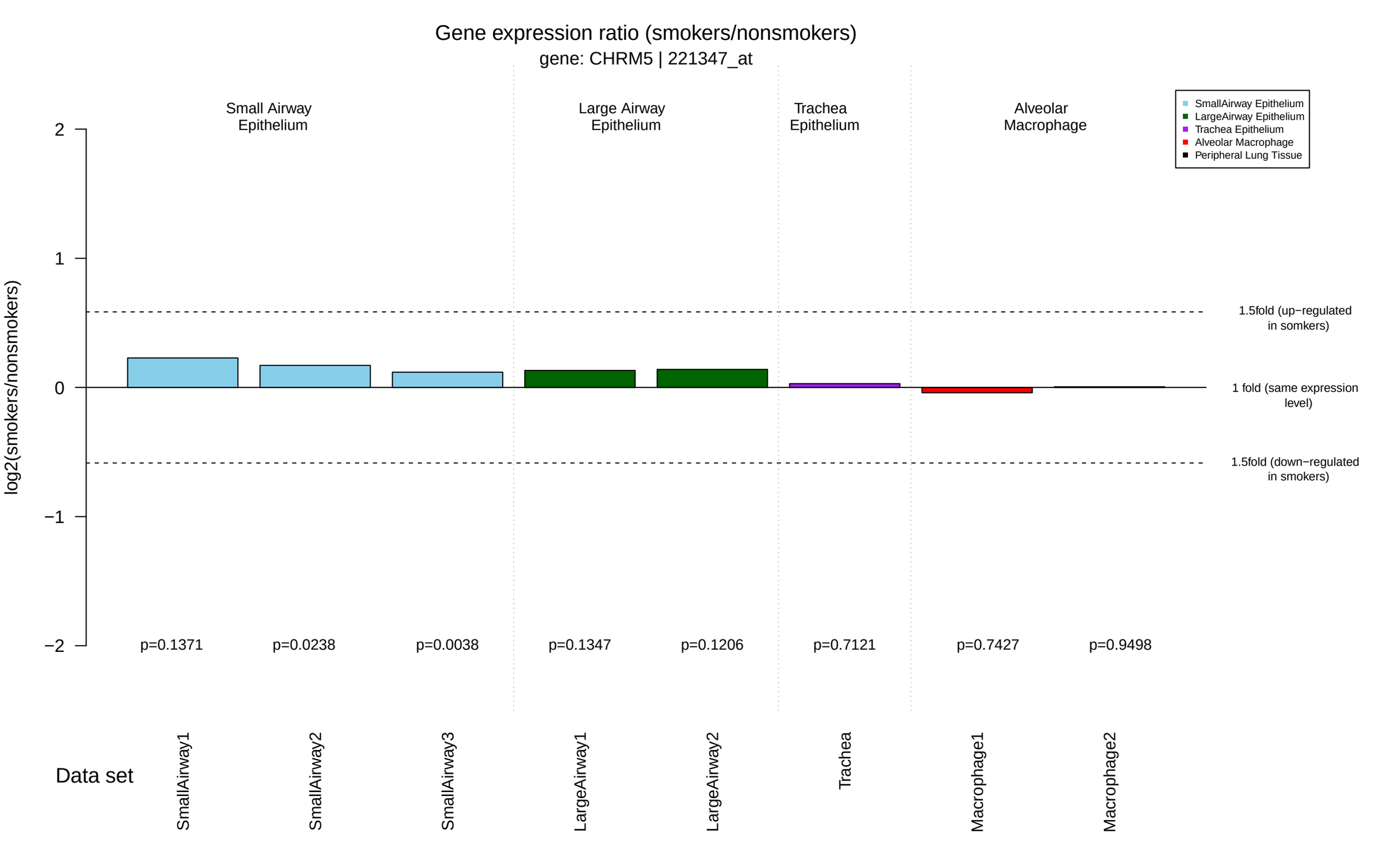
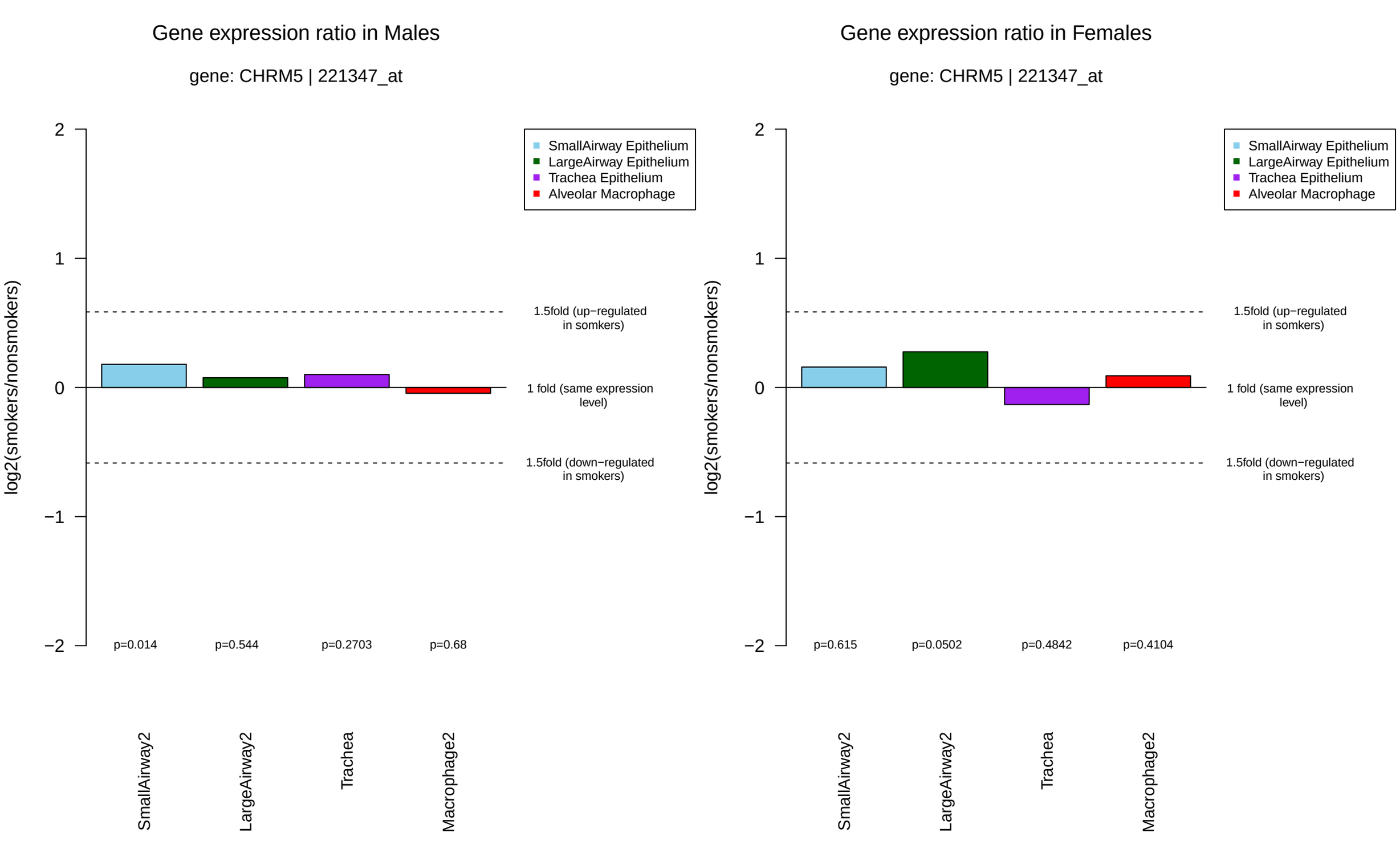
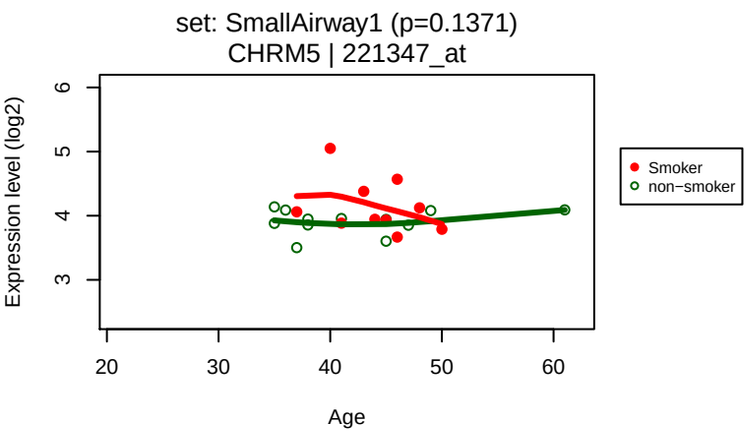
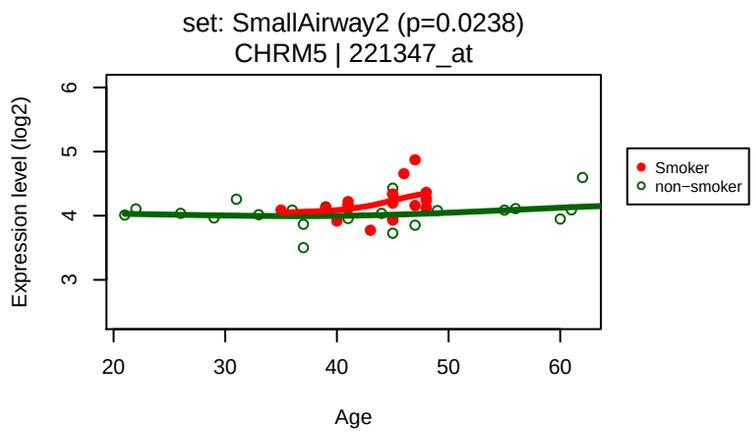
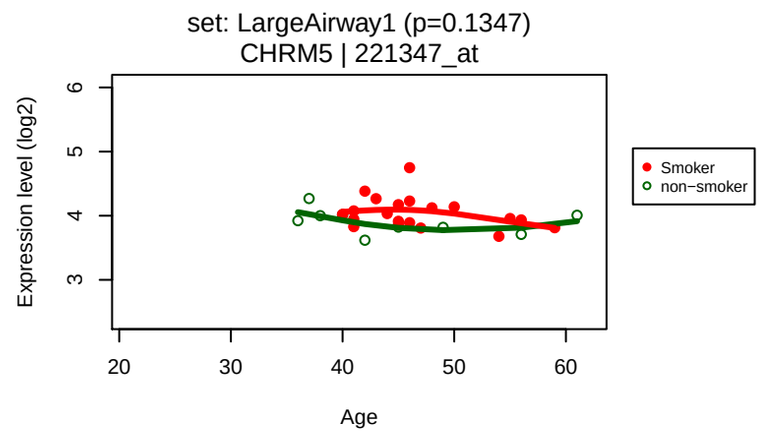

Gene expression of smokers and nonsmokers
gene: CHRM5 | 221347_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1371
p=0.0238
p=0.0038
p=0.1347
p=0.1206
p=0.7121
p=0.7427
p=0.9498
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

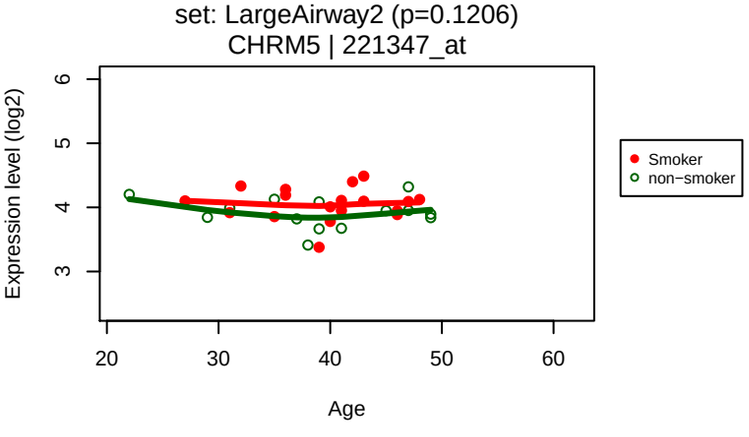

Gene expression ratio (smokers/nonsmokers)
gene: CHRM5 | 221347_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1371
p=0.0238
p=0.0038
p=0.1347
p=0.1206
p=0.7121
p=0.7427
p=0.9498
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

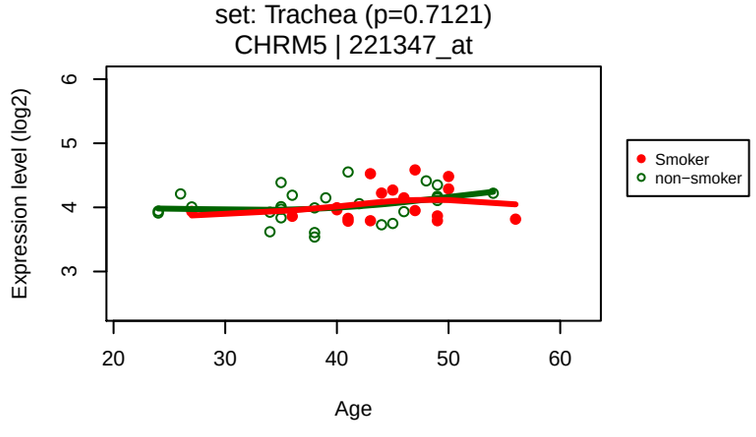

Gene expression ratio in Males
gene: CHRM5 | 221347_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.014
p=0.544
p=0.2703
p=0.68
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRM5 | 221347_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.615
p=0.0502
p=0.4842
p=0.4104
SmallAirway2
LargeAirway2
Trachea
Macrophage2

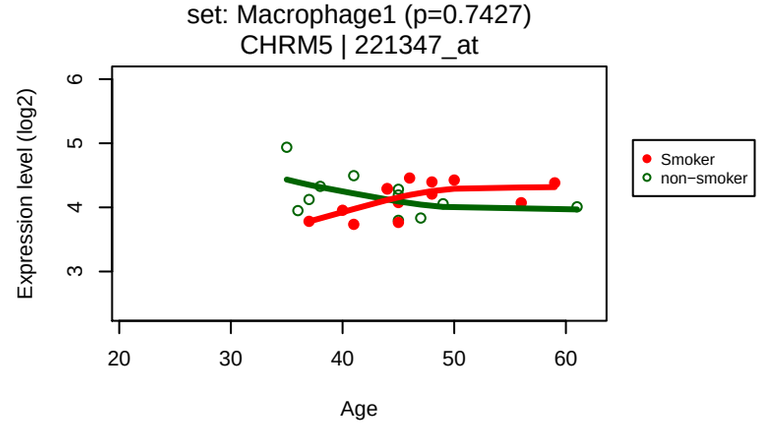

set: SmallAirway1 (p=0.1371)
CHRM5 | 221347_at
Expression level (log2)
Age
Smoker
non−smoker

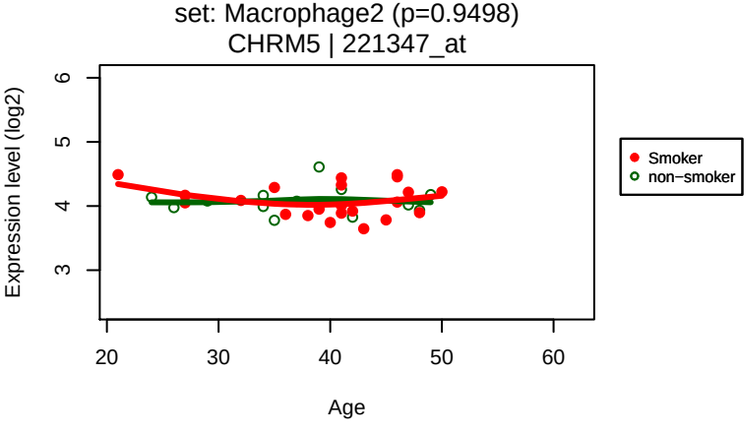

set: SmallAirway2 (p=0.0238)
CHRM5 | 221347_at
Expression level (log2)
Age
Smoker
non−smoker

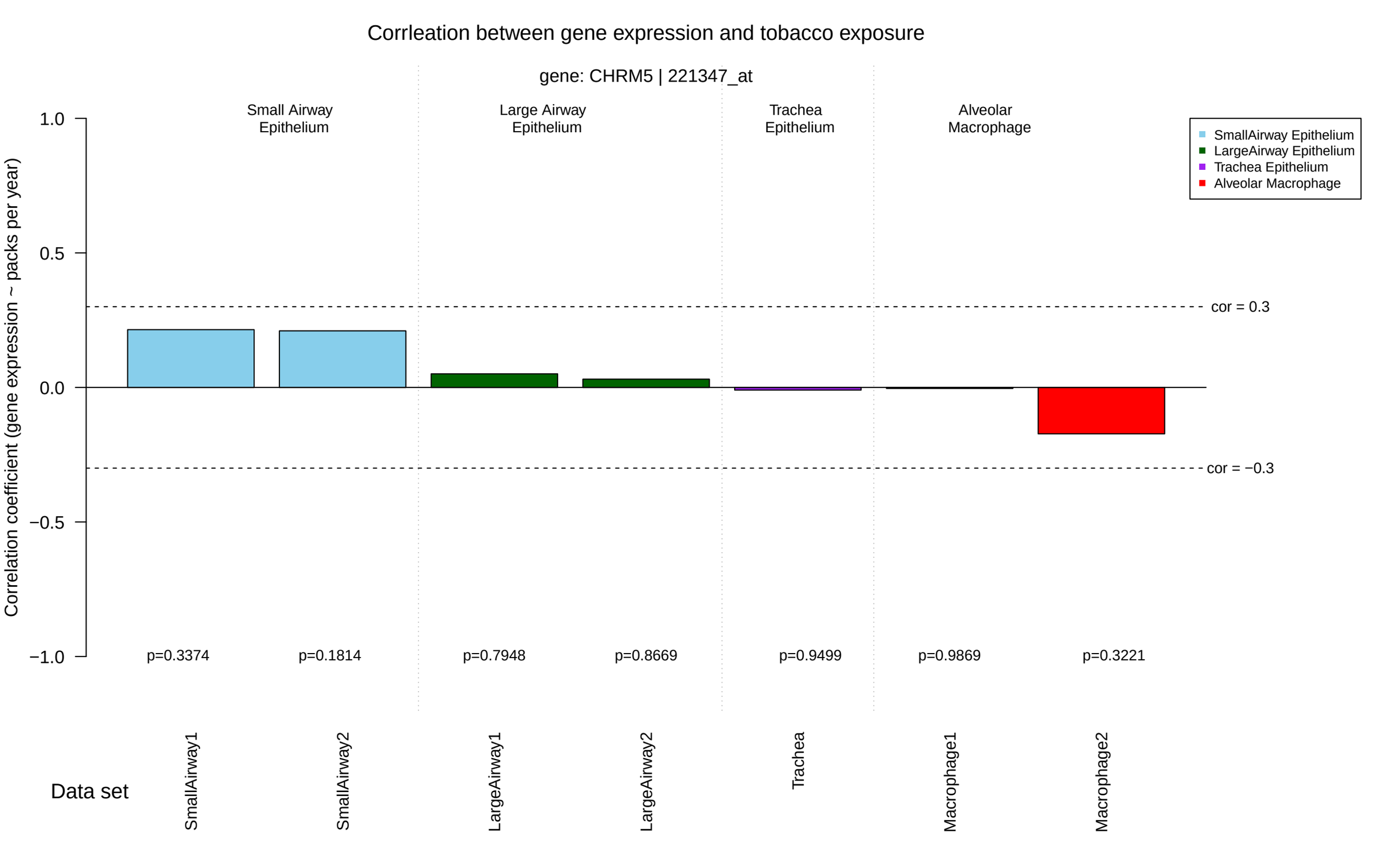

set: LargeAirway1 (p=0.1347)
CHRM5 | 221347_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1206)
CHRM5 | 221347_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7121)
CHRM5 | 221347_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7427)
CHRM5 | 221347_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9498)
CHRM5 | 221347_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRM5 | 221347_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3374
p=0.1814
p=0.7948
p=0.8669
p=0.9499
p=0.9869
p=0.3221
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2